

Modern To Do List

Alex Carpenter

Colorado State University Global

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Dr. Farhad Bari

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The 'Modern Photo Gallery' application is intended to be a simple interface for capturing and displaying photographs. Jetpack Compose was primarily used to construct the user interface. Material Design 3 was used for things like the top application bar, buttons, cards, and floating action button. 'CameraX' was what provided the camera functionality for the application. 'CameraX' handles many differences between Android devices and connects camera operation to the application's lifecycle (CameraX Architecture, n.d.). When a photo is taken in the app, it is saved to an app-specific directory on the device and displayed in a grid for the user to view.

Coil was selected to load images efficiently within the Compose interface. Rather than manually decoding full-resolution image files into memory, Coil performs image loading asynchronously and includes features such as caching, image-size optimization, and automatic cancellation of unnecessary requests (Coil, n.d.). The gallery also uses a lazy grid so that Compose primarily creates and displays the items currently needed on the screen. The purpose of doing this was to help reduce unnecessary memory consumption and improve scrolling performance when the number of stored photographs increases.

User experience and privacy were also important considerations. The application requests camera access only after the user chooses to open the camera instead of requesting the permission immediately when the application launches. Android's runtime permission system allows the user to approve or reject camera access before the feature is used (Request App Permissions, n.d.). The application does not request general access to the user's existing photo library because the gallery only displays images that it creates. App-specific storage therefore reduces unnecessary permission requests while keeping the application's files separated from unrelated user photographs. The interface also provides clear navigation between the gallery and camera, immediate feedback while an image is being saved, and an adaptive image grid that can accommodate different Android screen sizes.

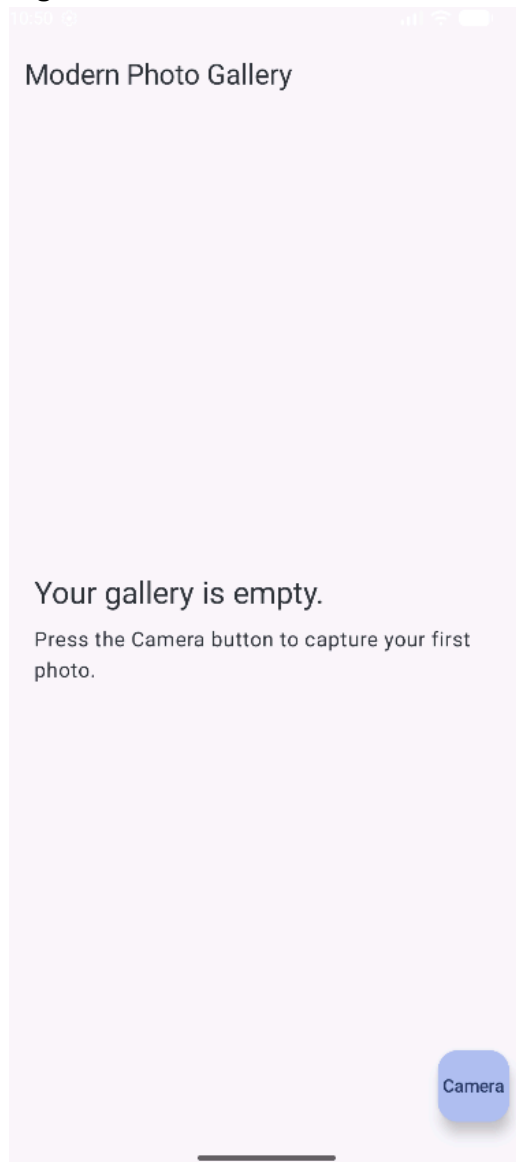
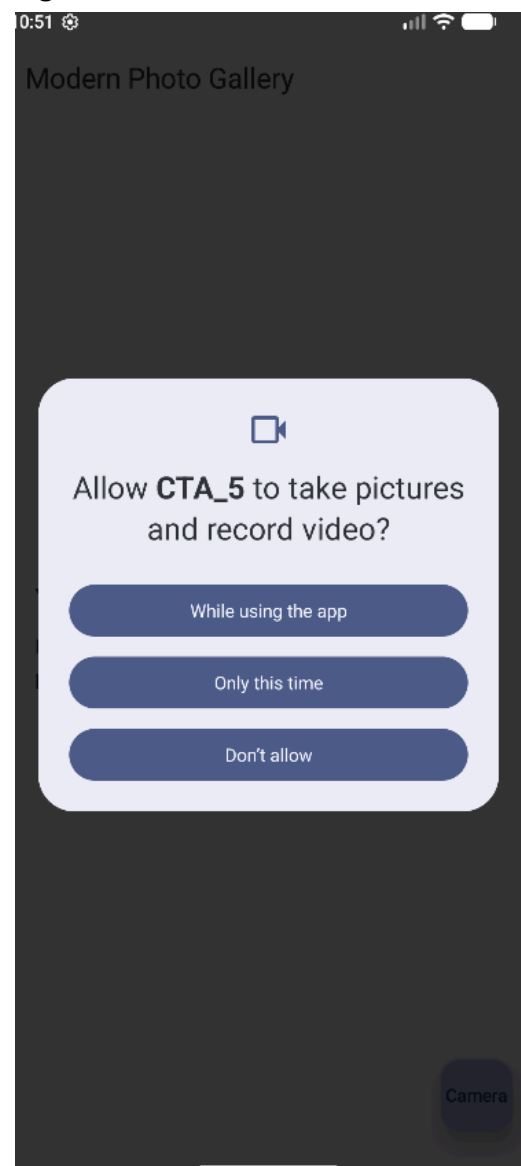
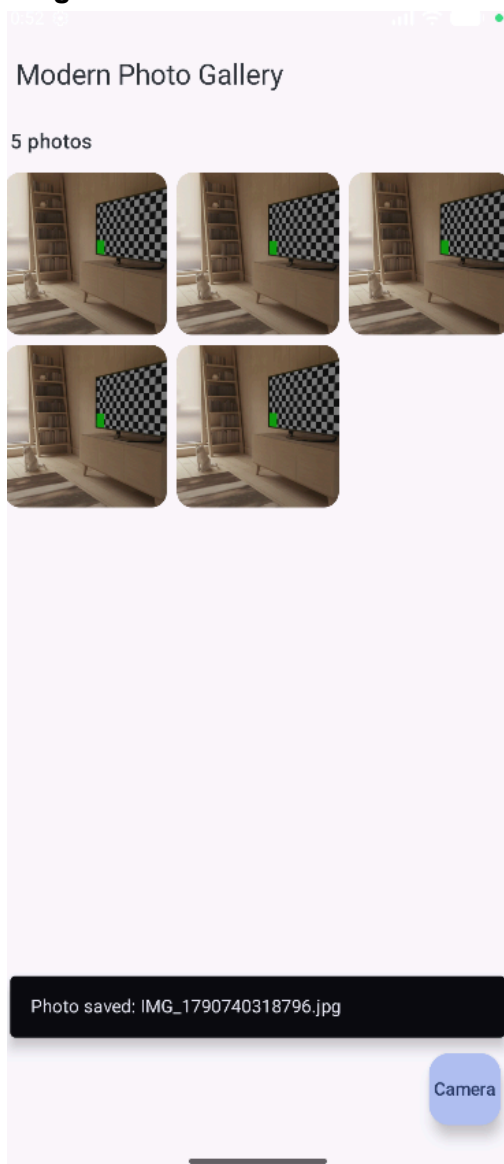
Figure 1.*The empty gallery screen.***Figure 2.***The app asks for permission to use the camera.*

Figure 3.

The camera is on in the app.

Figure 4.

The gallery grid being filled.

References

CameraX architecture. (n.d.). Android Developers. Retrieved September 29, 2026, from <https://developer.android.com/media/camera/camerax/architecture>

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Coil. (n.d.). Coil-Kt.Github.io. Retrieved September 29, 2026, from <https://coil-kt.github.io/coil/>